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Introduction:

Project title: Public Transport Service Frequency and Coverage in Metropolitan Victoria

Topic detail:
This project analyses the relationship between population growth, public transport coverage and frequency, and socioeconomic disadvantage across Statistical Area Level 2 (SA2) regions in metropolitan Victoria. It investigates how the current metropolitan public transport services support areas in Melbourne with historically higher population growth and demographic changes. Both temporal and spatial aspects of the transport system are considered, including analysis of differences in services across weekdays, weekends and times of day.

Problem Description:
The accessibility and frequency of public transport services in Victoria is crucial to support the ever increasing population of the state. Public transport is interwoven into the lives of Victorians, where about 80% of Victoria's population use public transport (Royal Automobile Club of Victoria, n.d.), making it important for all citizens and tourists. The system faces significant pressure to keep pace with the growing resident numbers, which can lead to inadequate service coverage and frequency. By researching the frequency and coverage of the current public transport system in relation to population and socioeconomic disadvantage, it can provide guidance to government agencies for transport planning and resource allocation when prioritising areas that require more attention.

Brief Motivation:
I recently moved to Melbourne and rely on public transport in my day-to-day life. I wanted to find out how accessible the public transport services were and if there was any connection between socio-economic status, service frequency, and area coverage that could aid in bridging gaps in social inequality and support more equitable resource allocation.

Questions:
1. Which statistical area level 2 (SA2) regions in metropolitan Victoria have historically had the greatest population growth, and how well do the current public transport services serve these areas?
2. How does public service coverage and frequency vary over different time periods (e.g. time of day/weekends) and how does this service coverage and frequency relate to historically more socioeconomically disadvantaged SA2 regions in metropolitan Victoria?

Data Wrangling and Checking:

Data Sources:

A. **Population estimates by SA2 and above, 2001 to 2024 (Table 1: Estimated resident population, Statistical Areas Level 2, Australia)**. The data provides annual estimated resident population for SA2s in Australia with records from 2001 to 2024. It is tabular data in XLSX format with ~2.5K rows x 34 columns. It contains both categorical attributes (State/Territory (S/T) code, S/T name, Greater Capital City Statistical Areas (GCCSA) code, GCCSA name, Statistical Areas Level 4 (SA4) code, SA4 name, Statistical Areas Level 3 (SA3) code, SA3 name, SA2 code, SA2 name) and numeric attributes (Yearly population estimates (2001 - 2024)). The data was used to identify high population growth SA2 regions in metropolitan Victoria between 2001 and 2024, providing the foundational demographic trend for comparing population growth with transport service supply. (https://www.abs.gov.au/statistics/people/population/regional-population/2023-24#data-downloads)

B. **General Transit Feed Specification (GTFS) Schedule**, 2026. The data contains detailed timetable information on all public transport services in Victoria structured in separate folders for each mode of public transport. It is relational tabular data with multiple TXT files per mode of transport describing the current 2026 public transport schedule. Key attributes include categorical (route_id, trip_id, stop_id, service_id, stop_name), spatial (stop_lat, stop_lon), and temporal (arrival_time, departure_time) types. It was first published in December 2024 and last updated in April 2026, but the data was downloaded in March 2026. The dataset enables evaluation of coverage and frequency of services in different SA2 regions and over different time periods (time of day/day of week). (https://discover.data.vic.gov.au/dataset/gtfs-schedule)

C. **Census Geopackage G33 - Total household income (weekly) by household composition**, 2021. The geopackage contains census data on the total weekly household income by household composition and different geographic layers for 16 Australian Statistical Geography Standard (ASGS) structures. It is a spatial dataset with ~500 rows x 64 columns covering the 2021 census data, released on 28 June 2022. It contains categorical attributes (SA2 code, SA2 name), polygon geometries, and numeric attributes (SA2 area, weekly household income by household composition (family/non-family/total)). The data provides spatial context, allowing spatial analysis of the relationship between transport access, population growth, and socioeconomic disadvantage at the SA2 level. (https://www.abs.gov.au/census/find-census-data/geopackages?release=2021&geography=VIC&gda=GDA2020&topic=HIHC)

D. **Statistical Area Level 2, Indexes, SEIFA 2021 (Table 1: Statistical Area Level 2 (SA2) SEIFA Summary, 2021).** The data contains information on the Socio-Economic Indicators for Area (SEIFA) for SA2 regions in Australia. It is tabular XLSX data with ~2.4K rows x 11 columns based on 2021 census data and released on 27 April 2023. It contains categorical attributes (SA2 code, SA2 name), numeric attributes (usual resident population, Index of Relative Socio-economic Disadvantage (IRSD) score, Index of Relative Socio-economic Advantage and Disadvantage (IRSAD) score, Index of Economic Resources (IER) score, Index of Education and Occupation (IEO) score), and ordinal attributes (IRSD decile, IRSAD decile, IER decile, IEO decile). The data was used to identify socioeconomically disadvantaged SA2 regions in metropolitan Victoria, allowing for analysis of service coverage and frequency across different levels of socioeconomic disadvantage. (https://www.abs.gov.au/statistics/people/people-and-communities/socio-economic-indexes-areas-seifa-australia/latest-release)

Data sources A, B, and C were used to address Question 1, while Question 2 was answered using data sources A, B, C, and D.

Data Wrangling and Data Checking:
Metropolitan trains, trams, and myki bus services GTFS schedule data was imported from separate folders and combined into single datasets. Relevant files (stops.txt, stop_times.txt, trips.txt, and calendar.txt) were read using read.delim() and merged using bind_rows(). During this merging process, inconsistencies in variable types were identified. An error appeared showing that stop_id and parent_stations were stored as different data types across the different modes of transport. This was verified using the typeof() function and resolved by explicitly converting both variables to character format using mutate() and as.character(), ensuring compatibility across datasets and preventing join errors. Unnecessary columns were dropped from the consolidated stops dataset, retaining only relevant attributes (stop_id, latitude, longitude) to improve efficiency. The stop_times, trips, and calendar tables were also similarly combined to create complete system-wide schedule datasets.

Services were filtered to represent typical weekday operating conditions by selecting only services operating on at least one weekday using the calendar dataset. This is because weekend services are usually less consistent, especially across outer metropolitan areas. This dataset was joined to trips and stop_times to produce a dataset of weekday stop events.

Population data was imported and cleaned by renaming relevant variables into consistent snake case format and filtered to SA2 regions within the Greater Melbourne GCCSA (metropolitan Victoria) areas. Absolute change in population (2001 - 2024) was derived by finding the difference between 2024 and 2001 population estimates and percentage change in population was calculated by dividing absolute change by the 2001 population estimate then multiplying by 100.

After calculating percentage change between 2001 and 2024 populations, I checked the distribution of these values using summary() and found that some SA2 regions had infinity, NaN, and extremely high (> 20000% change) values. Checking the 2001 and 2024 estimates for these records revealed that infinity values were caused by 0 population in 2001 (division by zero), the NaN values were caused by 0 population in both 2001 and 2024, and extremely high percentage change values were due to new/emerging SA2s with very small initial 2001 populations and high absolute population growth rates distorting the metric. These errors were resolved by restricting population change to SA2 regions with a 2021 population greater than 500, ensuring meaningful and stable comparisons while retaining all regions for absolute change analysis.

Using the sf package, spatial processing was conducted. The st_layers() function was used to find the layer name and coordinate reference system (CRS) name of the SA2 layer in the GeoPackage. The SA2 boundary data was imported using read_sf() and filtered to include relevant variables only (SA2 code, SA2 name, SA2 geometry) using select(). Only SA2s within metropolitan Victoria were retained by using left_join to join the population data to the SA2 boundary data. Stop locations were converted into spatial points using st_as_sf() with crs = 7844 (GDA2020) to align with the CRS in the SA2 boundary data.

The spatial join function st_join() was utilised to combine the metropolitan SA2 boundary data with the spatial stop location data, assigning each stop to a SA2 region with left = FALSE ensuring that only stops within Greater Melbourne were retained. Service intensity was computed by joining the stop times to the SA2-coded stop location data and counting the total number of stop events per SA2, providing a measure for operational service supply.

Before joining, the SA2 service intensity dataset had 357 rows, whereas, the population and SA2 boundary data with stop location assignment had 361 rows. After joining the datasets together with left_join(), summary() was used to check the distribution of values in the dataset, and *NA* service intensity values were identified. I realised that this was because of discrepancies in row counts prior to joining, revealing that some SA2s had no service intensity value due to absence of services. To resolve this, they were replaced with 0 to correctly represent absence of services instead of missing data. Service per capita was derived by dividing service intensity by 2024 population estimates. If the 2024 population was 0, the service intensity values were set to 0 to prevent undefined results.

Service frequency and coverage was analysed across different time of day periods (morning peak (6am - 10am), midday (10am - 2pm), interpeak (2pm - 4pm), afternoon peak (4pm - 6pm), and evening (6pm - 10pm)) based on departure times and days of the week (weekday, Saturday, Sunday). Any stop events outside these temporal ranges were excluded. A complete base dataset was created using expand_grid() to include all SA2, time period, and day combinations, preventing data loss during aggregation.

Service frequency was calculated as trips per hour. For weekdays, a weighted average approach was used, where a new variable was introduced into the calendar data representing the number of weekdays each service ran. Trip counts were weighted by this value and divided by 5, providing a true average weekday service frequency. For Saturday and Sunday frequencies, trips per hour was derived by counting the number of stop events per SA2 and time period before dividing by the duration of each time period. All frequency datasets were combined using bind_rows(), joined to the base dataset and checked using the summary() function. The summary() function showed that there were 95 SA2 regions with *NA* trips and trips per hour values; these values were due to absence of trips within those regions, so for completeness, these values were set to zero to represent absence of stops instead of missing data.

Service coverage was defined as the number of unique active stops per SA2 region. For weekdays, coverage was calculated by expanding services across individual weekdays using pivot_longer(), then identifying unique stop activity per SA2 and time period, before averaging across the 5 days to produce an average weekday coverage metric. For Saturday and Sunday, coverage was calculated by counting the number of distinct stop IDs per SA2 and time period. All coverage datasets were combined using bind_rows() and joined to the base data. Using the summary() function, it showed that there were 95 SA2 regions with *NA* active stop values; these values were due to the absence of active stops within those regions. For completeness, these values were set to zero to reflect absence of stops instead of missing data.

The 2021 SEIFA data was imported and cleaned by renaming relevant variables to consistent snake case format and converting data types to ensure consistency across datasets. The IRSD score was extracted and joined to the SA2 boundary data using SA2 codes. As small populations can significantly impact the reliability and sensitivity of IRSD scores resulting in more extreme values (Capuano, G., n.d.), the reliability of IRSD scores was improved by excluding SA2 regions with population estimates of less than 1000 residents in 2021.

All analysis was conducted using R and R studio, with the tidyverse, sf, and readxl packages for efficient data manipulation and analysis. Tableau was used for visualisations.

Data Exploration:

The data exploration process focused on understanding spatial and temporal patterns in the public transport data across metropolitan Victoria, and how these patterns related to population growth and socioeconomic disadvantage. The wrangling process combined 4 datasets: GTFS public transport schedules, SA2 population estimates, SA2 boundary geometries, and SEIFA socioeconomic indexes.

Exploration of these datasets used tools such as R and RStudio with tidyverse, ggplot2, sf, and readxl libraries due to its strong capacity for data manipulation and wrangling. The exploration process followed three main stages:

1. Transport service intensity and spatial distribution
2. Temporal variation in service frequency and coverage
3. Socioeconomic context with SEIFA integration

All variables were aggregated at the SA2 level to enable consistent spatial comparison across datasets.

Question 1:
Population change was measured using both absolute change (difference between 2001 and 2024 population estimates) and percentage change, calculated for SA2 regions with a 2001 population greater than 500 to avoid distortion due to small baseline values. Summary statistics using summary() indicated that absolute population change was heavily right-skewed, with a mean of 5126, median of 3159, and a range from -2358 to 34498, suggesting the potential presence of extreme high-value outliers. Percentage change showed even more skewness, with a mean of 152.85%, median of 25.54%, and a range from -17.24% to 3951.66%. The large gaps between mean and median in both measures indicate that some SA2 regions experienced extremely high population growth.

Multiple visualisations were used to capture both spatial patterns and magnitude differences in population change. A choropleth map of absolute change highlights geographic growth distribution, while ranked bar charts provide a clear distinction of the highest growth regions. The combination of these visualisations provides a more comprehensive understanding of population change and avoids relying on a single visualisation to convey the patterns.

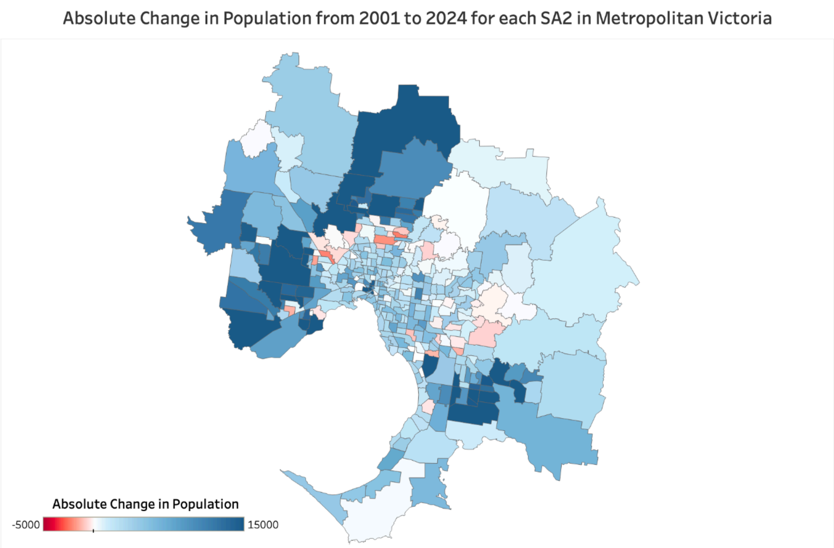

Figure 1. Absolute change in population between 2001 and 2024 for SA2s in metropolitan Victoria

The choropleth map uses a diverging colour scale (red - white - blue) centred at zero, where red indicates population decline and blue indicates population growth, allowing for easy identification of population change clusters. Colour limits were set from -5000 to 15000 to account for skewness, preserving variation and preventing extreme growth values from compressing the colour distribution. The map shows that the highest growth regions (dark blue) are concentrated in outer metropolitan areas, suggesting urban expansion and new housing development between 2001 and 2024. In contrast, central SA2 regions showed slower growth or slight decline (light blue/red) likely reflecting density limits, higher housing costs, or demographic changes.

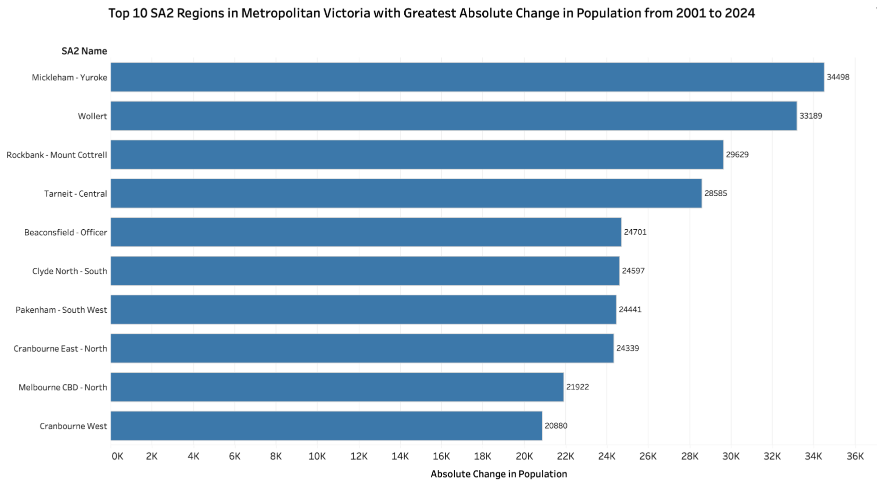

Figure 2. Top 10 SA2 regions in metropolitan Victoria with greatest absolute population change between 2001 and 2024

To complement the map visualisation, the top 10 SA2s by absolute population change were visualised using a ranked bar chart, allowing for precise comparison of growth magnitudes, which is less clear on the map. The x-axis represents the absolute change in population between 2001 and 2024, while the y-axis represents the SA2 name. The visualisation shows that Mickleham - Yuroke experienced the greatest absolute increase, with a growth of 34498 residents.

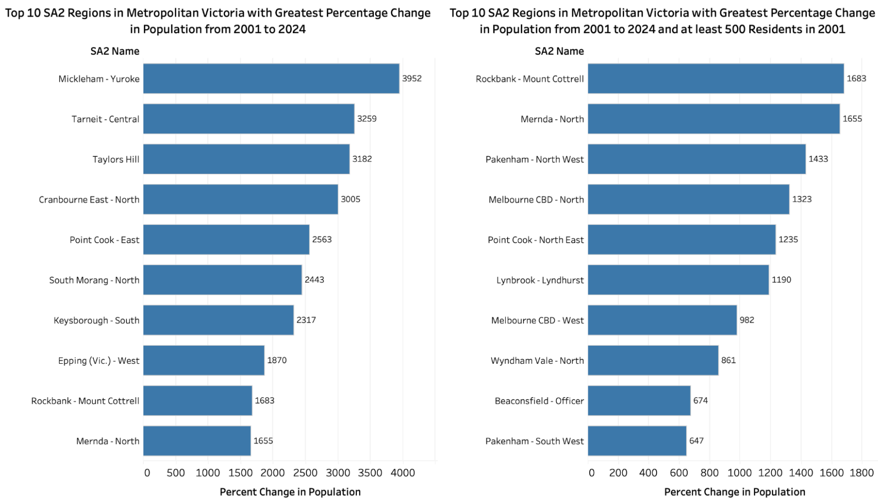

Figure 3. Top 10 SA2 regions in metropolitan Victoria with greatest percentage population change between 2001 and 2004 (left plot: for all regions, right plot: regions with at least 500 residents in 2001)

Percentage change was also analysed to account for relative growth. However, this measure is quite variable and sensitive to small baseline populations, so SA2 regions with less than 500 residents in 2001 were excluded in a second comparison. The x-axis represents the percentage change in population between 2001 and 2024, while the y-axis represents the SA2 name. When all regions are included (left plot), Mickleham - Yuroke appears as the highest relative growth region. However, when excluding SA2s with small 2001 populations (right plot), Rockbank - Mount Cottrell is the greatest relative population growth region, demonstrating the importance of controlling baseline population when interpreting percentage change.

Spatial distribution of transport service supply was measured using service intensity, defined as the total number of scheduled stop events within each SA2 region during a typical week (weekdays only), reflecting the total supply of scheduled public transport services across metropolitan Victoria, rather than total number of physical stop infrastructures. Using summary(), the summary statistics for service per capita indicated a right-skewed distribution, with values ranging from 0 to 292.6429, a median of 1.2290, and a mean 3.4975. The large difference between the mean and median values suggests the presence of potential high-value outliers in the data, likely driven by areas with relatively small populations and high service activity. To analyse the service supply for high-growth regions, the top 10 SA2 regions from each population change metric (absolute, percentage, percentage with a 2001 population threshold of 500) were selected from the main dataset, and plotted in a scatterplot. This is an appropriate choice of visualisation because it enables direct comparison between the two numeric variables, facilitating easy identification of patterns or trends.

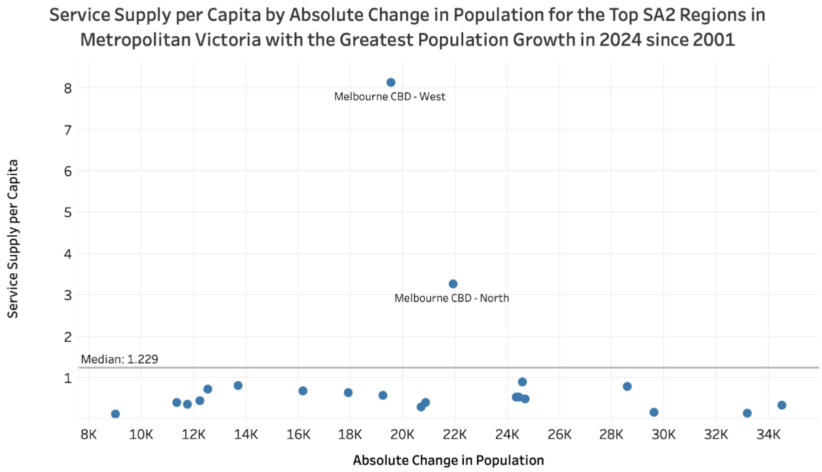

Figure 4. Service supply per capita by absolute population change for the top high-growth SA2 regions in metropolitan Victoria

The x-axis represents absolute population change (2001-2024), while the y-axis represents service supply per capita. A horizontal reference line represents the median service supply per capita across all SA2s in metropolitan Victoria, providing a benchmark for service supply. The plot shows that most high-growth regions fall below this median line, suggesting that there is a shortage in service supply for the populations of these regions. Only 2 high-growth SA2 regions (Melbourne CBD - West and Melbourne CBD - North) had service supply per capita values above the median, reflecting central transport hubs with disproportionately high service levels relative to residential population.

Question 2:

Public transport services were split into five different time of day periods for temporal analysis: morning (6am - 10am), midday (10am - 2pm), interpeak (2pm - 4pm), afternoon (4pm - 6pm), and evening (6pm - 10pm). Service frequency was defined as the number of trips per hour per SA2, with separate analyses conducted for weekday, Saturday, and Sunday services. Summary statistics showed that the distribution of trips per hour was right-skewed, with a median of 331.2, mean of 467.5, and a range from 0 to 6169.1. The large gap between median and mean suggests the presence of high-value outliers, likely reflecting highly serviced central areas. Statistical ANOVA tests using summary() and aov() were conducted on visualised variables, allowing for comparison between mean values across different times of day and days of the week, further supporting the findings from the bar chart visualisations.

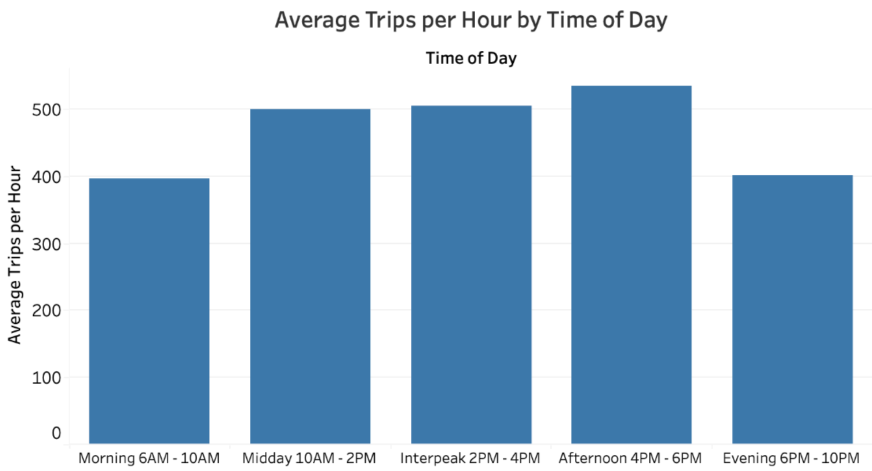

Figure 5. Average trips per hour by time of day

A bar chart is an appropriate visualisation choice because the time of day variable is ordinal while the average trips per hour is numeric, allowing comparison of averages across different time periods and providing a clear, intuitive representation of relative differences across different time of day periods.

The x-axis and length of the bar/position on the y-axis represent the time of day period and average trips per hour, respectively. The bar chart shows a clear temporal pattern in service frequency, where afternoon has the greatest service frequency, reflective of peak post-work congestion demand, while midday and interpeak periods also maintain relatively high service frequency levels. Morning services have moderate frequency, which is lower than expected compared to the afternoon period and evening service frequency experiences a notable decline, consistent with reduced demand outside of peak commuting hours. The ANOVA test results were statistically significant (p = 7.91e-16), confirming that service frequency varies meaningfully throughout the day.

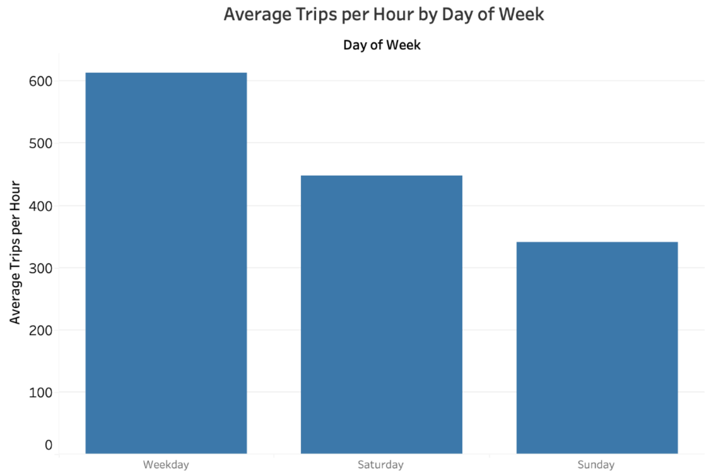

Figure 6. Average trips per hour by day of the week

Similarly, service frequency was also compared across days of the week using a bar chart with the length of the bar/position on the y-axis representing the number of average trips per hour and the x-axis representing the day of the week. The visualisation shows a clear hierarchy, with weekdays having the highest average frequency (about 600+ trips per hour), followed by Saturdays with a noticeable reduction in service frequency (about 450 trips per hour), and Sundays with the lowest service frequency (about 340 trips per hour). This pattern reflects public transport scheduling priorities aligned with weekday commuting demand (for work, school, and etc.) and lower, more dispersed weekend service demand. The statistical ANOVA test produced a p-value of <2e-16 which is less than 0.05, confirming that there is a statistically significant difference between average trips per hour on different days of the week.

Public transport service coverage was defined as the number of unique stops with at least one service during a given time period within each SA2, measuring spatial reach. Using summary() to check the summary statistics showed that active stops were slightly right-skewed, with a median of 52, mean of 54.46, and a range from 0 to 198, indicating relatively consistent coverage across regions with some high-value outliers.

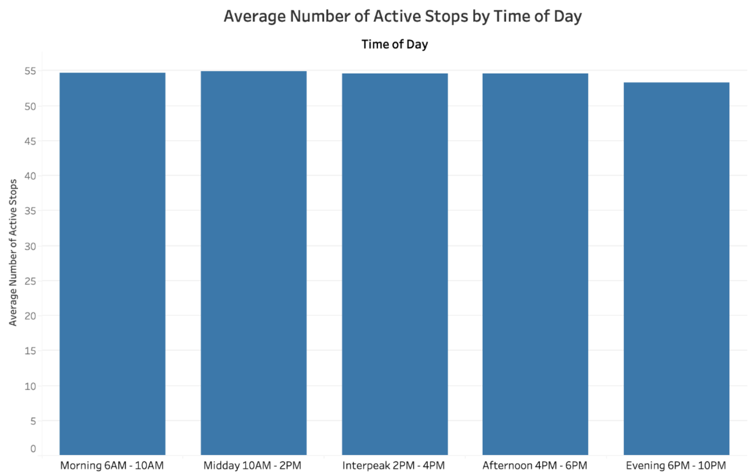

Figure 7. Average number of active stops by time of day

A bar chart was chosen to visualise this data because the data contains an ordinal time of day variable and a numeric average number of active stops variable. The bar chart allows for effective identification of differences in public service coverage across different time of day periods. On the bar chart, the x-axis represents the time of day, while the length of the bar/position on the y-axis represents the average number of active stops. It shows that there is minimal variability in service coverage during different times of the day, with only a slight drop in active stops in the evening, likely reflecting the adjournment of lower demand routes and areas later in the day. This suggests that while service frequency changes throughout the day, spatial network coverage remains relatively stable. The statistical ANOVA test produced a p-value of 0.73 which is greater than 0.05, confirming that differences across different times of day are not statistically significant.

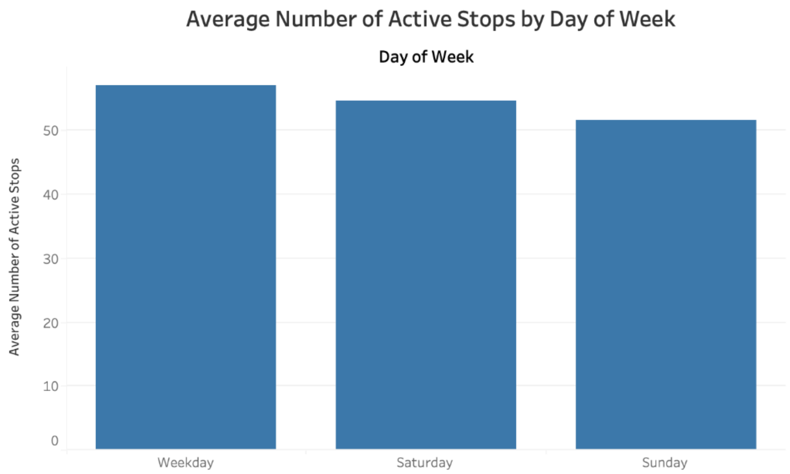

Figure 8. Average number of active stops by day of week

A bar chart is an appropriate choice of visualisation to display this data because the goal is to compare magnitudes of average number of active stops across categorical day of the week data. The bar chart provides clear visual identification of differences in service coverage between the day of the week groups. The x-axis represents the day of the week, while the position on the y-axis/length of the bar represents the magnitude of the average number of active stops. The bar chart shows a clear pattern where weekdays have more operating stops compared to the weekend, with Sunday having the lowest number of active stops reflecting the lower travel demand on weekends. However, these differences between active stop counts are relatively small. The statistical ANOVA test produced a p-value of 1.47e-07 which is less than 0.05, confirming that the variation between different days of the week is statistically significant.

SEIFA IRSD scores were joined to metropolitan Victoria SA2 boundaries, lower scores indicate more socioeconomic disadvantage and higher scores indicate less socioeconomic disadvantage.

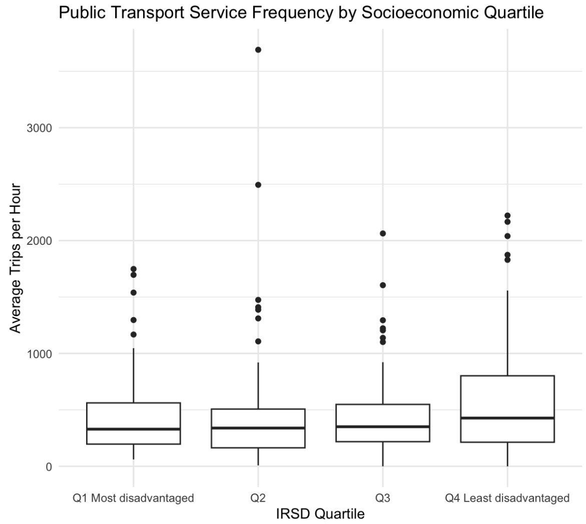

Figure 9. Public transport service frequency by socioeconomic quartile (IRSD score)

A boxplot is an appropriate choice of visualisation to compare service frequency (average trips per hour) across IRSD quartiles because it effectively summarises distribution through summary statistics (median, interquartile range, and spread) for each quartile, enabling easy identification of outliers and clear comparison between quartiles. The results suggest there is a weak positive relationship between socioeconomic status and service frequency with median service frequency slightly increasing as socioeconomic advantage increases. However, the interquartile ranges of the quartiles have substantial overlap indicating that differences between IRSD quartiles are small relative to variability within quartiles, suggesting that the relationship between socioeconomic status and service frequency is weak. There is significant variability within quartiles, with notable outliers present, particularly in quartiles 2 and 3. These outliers are associated with central urban SA2 regions such as Melbourne CBD - West, where there is high service frequency relative to their small residential population resulting in disproportionately large frequency values compared to other regions within the same IRSD quartile, highlighting an important limitation of the metric, which can be inflated in areas with low population but high transport demand, such as central business districts. Despite these outliers, the overall trend suggests that there is limited evidence of inequitable distribution of service frequency across socioeconomic groups. An ANOVA statistical test was conducted using summary() and aov() as it allows for comparison between mean values across different IRSD quartiles to further analyse and support the above findings. The statistical test produced a p-value of 0.106 which is greater than 0.05, indicating that there is no statistically significant difference in IRSD quartiles and any observed differences from the boxplot are likely due to random variation rather than meaningful socioeconomic disparities.

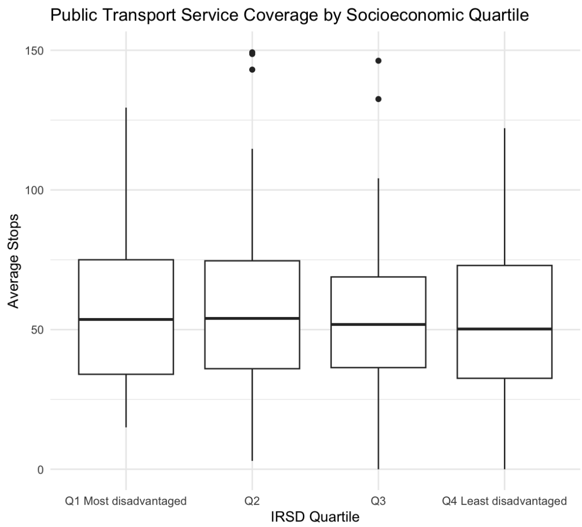

Figure 10. Public transport service coverage by socioeconomic quartile (IRSD score)

The position on the y-axis encodes the average number of stops, allowing for direct comparison of stop distribution levels across socioeconomic groups while, the x-axis uses ordered categorical position to represent IRSD quartiles of socioeconomic disadvantage (Q1-Q4), supporting meaningful ranking from most to least disadvantaged. The boxplot is a suitable visualisation for this data because it allows variability to be shown through interquartile range effectively showing the spread of the data without clutter. The median highlighted by the bold horizontal line inside each box emphasises central tendencies and outliers are represented by individual points beyond the whiskers, drawing attention to irregular service coverage values. Overall, the combination of position, length, and visual grouping enables efficient comparison of distribution, variability, and central tendencies across the four socioeconomic groups. The boxplot shows that distribution of active stops across all quartiles are very similar, with near identical medians, significant interquartile range overlap, and no clear trend across the socioeconomic groups. This suggests that public transport coverage is relatively evenly distributed across metropolitan SA2s, regardless of socioeconomic disadvantage. There are some outliers observed with more than 140 active stops, such as Preston - East, Oakleigh - Huntingdale, and Box Hill SA2 regions for Q2 and Point Nepean and Berwick - North for Q3. The outliers likely reflect specific urban characteristics, including transport hubs or larger geographic areas requiring more stops. An ANOVA test using summary() and aov() was conducted, enabling comparison between mean values across different IRSD quartiles and to support the findings from the boxplot visualisation.

The statistical test produced a p-value of 0.605 which is greater than 0.05 confirming that differences across IRSD quartiles are not statistically significant.

Conclusion:

The analysis investigates how population growth and public transport service supply vary across different SA2 regions in metropolitan Victoria, and whether the service distribution aligns with population growth and socioeconomic needs. The obtained results showed that population change between 2001 and 2024 had a highly uneven right-skewed distribution with significant growth concentrated in outer metropolitan areas. These regions experienced the largest absolute population increases, likely reflecting urban expansion and new housing developments, while some established inner areas showed slower growth or decline. When comparing population growth with service supply, a shortage of supply was evident. A small number of high-growth SA2s (Melbourne CBD areas) exhibited high service supply per capita, however, most high-growth regions fell below the median, suggesting that public transport supply has not matched population growth in many outer suburban areas, which could potentially contribute to accessibility challenges and increased car dependence. Temporal analysis showed clear, expected patterns in service frequency, with peak services occurring in the afternoon and on weekdays, reflecting typical commuting demand. In contrast, service coverage remained relatively consistent across different time periods, indicating that while frequency fluctuates, the spatial extent of the network is still very consistent. Differences across days of the week were statistically significant for both frequency and coverage, confirming reduced services on weekends. The socioeconomic analysis revealed a weak relationship between IRSD score and service frequency, with no statistically significant differences across quartiles. Similarly, service coverage was relatively uniform across socioeconomic quartiles. These findings suggest that variations in levels of service is more closely linked to urban structure and land use (e.g. CBDs and transport hubs) than to equity-based allocation. Overall, the analysis answered the research questions by showing that while population growth is spatially concentrated, transport service supply does not consistently align with this growth, and there is limited evidence of inequitable public transport service distribution based on socioeconomic disadvantage.

Reflection:

The project reinforced key concepts about effective data analysis and visualisation. It showed that a single visualisation alone is not always sufficient to capture complex spatial and statistical patterns and a combination of visualisations enable different dimensions in the data to be explored resulting in more thorough analysis and understanding. It also showcased that the choice of visual variables greatly influenced clarity and interpretability. As many variables were highly skewed, thoughtful scaling was required to avoid misleading representations, highlighting the importance of analysing data distributions. Without adjusting, extreme values would have dominated the visuals and concealed meaningful patterns. Service per capita and percentage population change metrics were easily distorted by small denominators while high service supply in central areas appeared to reflect commuter demand rather than resident needs, emphasising the importance of accompanying visualisations with contextual understanding. In hindsight, incorporating accessibility measures could have strengthened the analysis and interactive visualisations may have improved efficiency and effectiveness for pattern detection in the data. Further statistical modelling could have also provided deeper insights into the key drivers of public transport supply patterns.

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I acknowledge the use of generative AI tools to aid in explaining unfamiliar concepts but I take full responsibility for the content and its reliability. The following prompts were used and the material was cross checked with online sources as well.

- Do weekends generally have less public transport services than weekdays?
- Does Flemington Racecourse station only run for special events?
- Do SA2 regions in the Macedon Ranges use any of the Victorian metropolitan public transport services?
- What does IRSD score measure?
- Do small populations affect IRSD scores?